Medicine in Novel Technology and Devices 18 (2023) 100237

Contents lists available at ScienceDirect

Medicine in Novel Technology and Devices

journal homepage: www.journals.elsevier.com/medicine-in-novel-technology-and-devices/

Detection of anemia using conjunctiva images: A smartphone application approach

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A R T I C L E I N F O

Keywords:
Anemia detection
Pallor analysis
Conjunctiva
Machine learning
Red blood cells

A B S T R A C T

Anemia is one of the public health issues that affect children and pregnant women globally. Anemia occurs when the level of red blood cells within the body is reduced. Detecting anemia requires expert blood draw for clinical analysis of hemoglobin quantity. Although this standard method is accurate, it is costive and consumes enough time, unlike the non-invasive approach which is cost-effective and takes less time.

This study focused on pallor analysis and used images of the conjunctiva of the eyes to detect anemia using machine learning techniques. This study used a publicly available dataset of 710 images of the conjunctiva of the eyes acquired with a unique tool that eliminates any interference from ambient light. We combined Convolutional Neural Networks, Logistic Regression, and Gaussian Blur algorithm to develop a conjunctiva detection model and an anemia detection model which runs on a Fast API server connected to a frontend mobile app built with React Native.

The developed model was embedded into a smartphone application that can detect anemia by capturing and processing a patient's conjunctiva with a sensitivity of 90%, a specificity of 95%, and an accuracy of 92.50% on average performance in about 50 s.

1. Introduction

One of the public health problems affecting children and pregnant women universally is anemia. Anemia occurs when the body's supply of red blood cells is diminished [1]. Anemia is a health condition characterized by low blood hemoglobin levels due to the lack of enough red blood cells in the blood, which reduces the ability of the blood to transport oxygen [2]. Chaparro & Suchdev [3] stated that anemia affects about a third of the world's population and contributes to increased morbidity, mortality, decreased work productivity, and impaired neurological development. Wemakor [4], indicated that anemia has a high prevalence in low-income countries like Ghana especially in pregnant women, children, and elderly people. Nutritional factors such as iron deficiency, vitamins and minerals, and infectious diseases such as malaria, intestinal parasites, or genetic factors (hemoglobinopathies) are the main causes of anemia [5,6].

In many parts of the world, anemia is detected by using invasive laboratory techniques that require the extraction of blood by certified biomedical scientists or licensed laboratory technicians [7]. However, this invasive approach is costly, consumes more time, and is stressful for patients with chronic anemia, conditions that require frequent monitoring [8]. This (invasive approach) may introduce health professionals to blood-transmitted diseases or infections [5].

In low-income regions, especially where health facilities, equipment, and personnel are limited, the immobility of the standard clinical methods poses lots of inconvenience to patients as they sometimes travel from faraway places to access laboratories.

This has prompted the need for better alternatives to the existing methods of anemia diagnosis and thus the conduction of several studies for finding non-invasive methods by the scientific community over the years. Mannino et al. [5] stated that anemia may qualitatively correlate with the subjective assessment of pallor in fingernail beds, conjunctiva, and palmar creases of a patient's body. These regions of the body do not contain melanin and have a high affinity for bilirubin [9,10].

This study aimed to develop and implement a smartphone application that focuses on the analysis of the conjunctival colour of the eyes. This is

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https://doi.org/10.1016/j.medntd.2023.100237
Received 3 February 2023; Received in revised form 20 April 2023; Accepted 4 May 2023

a cross-platform mobile application that detects anemia using conjunctival photos (non-invasive approach) It provides easy access to anemia diagnosis mainly for people in remote areas where access to health facilities and personnel are limited.

This study used a positivism philosophy. This is because quantitative methods, such as statistical analysis, are recommended for investigating the quantitative link between conjunctiva pallor and Hb levels in determining anemia [11]. Positivist research gives trustworthy, consistent, and unbiased results [12].

In this study, a cross-platform mobile application that runs on the Android and iOS operating systems was developed with React Native in the Visual Studio Code Integrated Development Environment (VS Code IDE), with the use of conjunctiva of the eye images. This app act as the user interface or front end through which the user will communicate with our Anemia Detection Model. The user launches the app and uses their phone's camera to take pictures of the eye. The image is transferred to the FastAPI server, which executes our analysis model, over the phone's internet connection. The image processing takes place on a Python-based FastAPI server.

2. Related works

Several studies have been conducted in recent years to find alternative solutions to improve existing methods used in the diagnosis of medical diseases. A non-invasive method for detecting anemia with the use of a multilinear regression algorithm with the application of MATLAB using images of the conjunctiva of the eyes [5] which was incorporated into mobile apps.

Jiang et al. [13] developed HemaApp that connected led lights to a smartphone camera and a machine learning algorithm to measure the value of Hb as a non-invasive means of detecting anemia. The system measures Hb values through analysis of the blood at the user's fingertip and measures the absorption properties of the blood at different wavelengths of light. By analyzing the absorption pattern across multiple wavelengths of light, an estimation of hemoglobin is achieved using the machine-learning algorithm. Using fiber optics and a specific arrangement of light-emitting diodes, Kumar et al. [7] created a novel multi-wavelength spectrophotometry sensing platform to detect anemia. The study proposed a method where incident light intensity can be measured along with attenuated light intensity to support hemoglobin estimation.

A computer-based method that predicted anemia by analyzing images of the palpable conjunctiva was proposed [14]. Images captured from a smartphone camera are sent to UPCH servers for processing to predict anemia and the results are returned to the smartphone. Automatic segmentation is performed to extract the conjunctiva using CNNs [15], and then R and G components of the Erythema Index (EI) are calculated. Using these values and a neural network regressor, the prediction of anemia is estimated.

Mannino et al. [5] created a comprehensive non-invasive paradigm, on-demand diagnostics, which substitutes typical blood-based laboratory tests with merely a smartphone app and images. The research was designed to identify anemia. The app developed measures hemoglobin levels and identifies anemia by analyzing the colour and metadata of fingernail bed smartphone photographs.

Segmentation of the conjunctiva region for non-invasive anemia detection applications using deep learning was achieved [16]. Their proposed U-Net Based Conjunctiva Segmentation Model (UNBCSM) uses fine-tuned U-Net architecture for effective semantic segmentation of conjunctiva from the digital images of the eye captured by consumer-grade cameras in an uncontrolled environment. Their study proposed that their system be used as a preprocessing step for existing works to replace manual cropping of the conjunctiva region.

A computerized system was proposed by Refs. [17,18] to diagnose anemia using a modified Kalman filter (KF) and a regression method. They achieved this by computing the mean value of the red (R) component of the palpebral conjunctiva image as their recognition feature and used a penalized regression algorithm to find a nonlinear curve that best fits the data of feature values and the corresponding levels of Hb concentration. With the fitting curve, a three-level risk assessment scheme (RES) was maintained to determine the anemia status of a given subject, as well as the performance of the proposed approach. Results after experimentation showed that with the reformulated KF, the variance of the data could be reduced, which can facilitate the pattern recognition task at a later stage.

Dimauro et al. [19] quantified and minimized the subjective nature of the examination of the palpebral conjunctiva and proposed an approach of diagnostic support and autonomous monitoring. Their estimated system was made up of a smartphone and software module, a macro lens, and a special spacer. The software application facilitates and supports the monitoring of anemia and the potential for direct medical consultation by instantly transferring all data to a physician. After imaging the region of interest, the palpebral conjunctiva was extracted using the Superpixel segmentation (SLIC) algorithm. A K-means algorithm was applied to the image for colour clustering to abstract the image from irrelevant details. The random over-sampling examples (ROSE) technique and the SMOTHE technique were also used to balance to better estimate the relationship between anemia and conjunctiva pallor [19].

Collings et al. [20] established the palpebral conjunctiva erythema index (EI), calculated from photos captured with a compact camera or mobile phone and correlated the images with hemoglobin predicts anemia better than clinical assessment. Their research compared with the palpable conjunctiva; the palpebral conjunctiva's EI showed a stronger correlation with hemoglobin concentration.

Tamir et al. [21] utilized MATLAB's image processing algorithm and RGB thresholding to detect anemia from conjunctiva images. The procedure involves using an Android application developed to take a photo of the conjunctival pallor of the eye using the camera of a smartphone with appropriate controls and inadequate lighting. The photograph was then uploaded to a computer using any available method, including the Internet. The RGB spectrum of the anterior conjunctival pallor was extracted from the images by a computer algorithm, which compares it with a predetermined threshold value to determine whether the person is anemic or not.

Furthermore, Kasiviswanathan et al. [16] used the region of interest (ROI) of the image of the eye where it was manually cropped out in various white balances and then used the Ridge Regression algorithm, which is a regularized linear regression model. This was applied to the cropped ROI to estimate the Hb level of a patient. The system's overall output included a correlation of basic details like weight, height, and body mass index (BMI) of the patient to their Hb level. The proposed model was trained and performed well with any consumer-grade camera irrespective of the image sensor, image resolution, and lighting conditions.

To achieve this, images taken by multiple smartphones with different resolutions were used for training. The image features were extracted from the white balance change and the images of the conjunctiva were also included in the training data to improve the model in all lighting conditions. Selected image features, Hb values, and basic details were considered as input to design this model.

Using the k-Nearest Neighbor (k-NN) classification algorithm Dimauro et al. [2] aimed at developing a device that is not expensive and simple to use to assess anemic conditions based on the image analysis of a specific conjunctiva region. The a* value of the conjunctiva areas and Hemoglobin values were correlated using image processing in the CIE L*a*b* (CIELAB) colour space. No sample selection based on participant gender or clinical circumstances was used to arrive at the results. A k-NN classifier was set up to evaluate whether a subject requires a transfusion or at the very least attention from a doctor to save needless blood sampling.

Kasiviswanathan et al. [16] used a fully automated segmentation approach based on graph partitioning that reveals conjunctival regions while maximizing the association between colour characteristics and hemoglobin concentration in the blood. The ROIs identified by the algorithm were quantitatively compared with ground truth using state-of-the-art metrics for similarity and PCC between the a* component of the CIE L*a*b* colour space and hemoglobin values. The automatically segmented region of interest was appropriate for diagnostic procedures that involved quantitative measurement of hemoglobin from exposed conjunctival tissues.

To evaluate the effect of skin complexion on the accuracy of pallor analysis in anemia detection in children, clinicians performed blind-independent physical examinations and compared their findings with HemoCue 301 hemoglobin estimates [22]. The results showed that anemia was found more easily in children with dark skin. The most sensitive areas were the conjunctivae and the palm pallor (78.6% and 69.2%, respectively). They concluded that clinical pallor is a useful screening tool for anemia, but it is not a diagnostic test because its sensitivity and specificity fluctuate depending on location and skin type. As a result, integrating information from any of the sites can help improve anemia detection in children.

Dimauro et al. [19] utilized an effective approach to automatically segment conjunctiva sections and obtain the highest possible correlation of Hb values in the blood with digital images where the palpebral conjunctiva was critical to the detection of anemia using conjunctival images. Using image processing in the CEILAB (CIE L*a*b*) colour space, a novel device for the acquisition of the conjunctiva of subjects was developed. A deep learning approach was utilized by Zhang et al. [23] to detect anemia with the use of 316 medical images Hb value of $<$13 g/dL in men and Hb $<$ 12 g/dL in women. The study achieved an accuracy of 68.42%, a specificity and a sensitivity of 83.33% and 61.53% respectively.

Ghosal et al. [24] offered a spectroscopy-based blood hemoglobin level monitoring model as a unique, autonomous, smart anemia-care solution for issues associated with invasive anemia detection techniques. A smartphone camera was used as a sensor; this approach calculated the hemoglobin level based on colour spectroscopy of the region of interest, which is the conjunctival pallor extracted. Anemia was diagnosed when the expected hemoglobin level is less than 11.5 g/dL.

Dimauro et al. [25] discussed non-invasive ways for monitoring and identifying possible risks of anemia, as well as smartphone-based devices to conduct this operation, and concluded that non-invasive approaches are promising in tackling medical disorders such as anemia detection with the application of machine learning models in this digital era. Hasan et al. [26] developed a novel smartphone-based non-invasive Hb level prediction model by analyzing the hue, saturation, and value (HSV) of a fingertip video using Python OpenCV, MATLAB, and R statistical tools as data analysis tools. To facilitate smartphone deployment, the proposed methodology estimated hemoglobin levels using image analysis techniques.

From the related works reviewed, most of the datasets participants' age group was not indicated in the studies, while in other studies the age group was based on adults as indicated in Table 1, which gives details of the dataset samples used in the related works reviewed and that of the proposed study. However, this study's dataset focuses on the conjunctiva of the eye images collected from children aged 6–59 months. In addition, the sample size of the dataset used for the proposed study is higher than the dataset used by other studies stated in the related works which are indicated in Table 1.

Table 1
Summary of dataset sample and participants' age group of some related works.

Reference	Dataset Size	Participant Age Group
Tamir et al. [21]	19	–
Peksi et al. [27]	20	–
Suner et al. [28]	344	19–96 years
Noor et al. [29]	104	–
Jain et al. [30]	99	–
Roychowdhury et al. [31]	27	–
Dimauro et al. [32]	211	20–88 years
Yadav & Singh [33],	400	–
Proposed Study	**710**	**6–59 months**

Moreover, this study focused on the development of a smartphone application to be hosted on the Google Play Store for Android users to access the application without a fee. This would go a long way to eradicate the challenges encountered during the invasive process of anemia diagnosis and treatment, as compared to the other studies whereby the models are only developed without its implementation.

3. Methodology

This study was segmented into three stages; the dataset was gathered and pre-processed, and the models were trained and validated with the dataset, with the application of cross-validation to avoid overfitting. Finally, we tested the models with real-time data (conjunctiva images) with the corresponding Hb values from the laboratory with statistical analysis.

3.1. Data collection method

To collect the dataset, we developed a data collection system and provided licensed medical laboratory personnel with training on how to utilize the "Kobo Collect" app. A form was created to gather patient biodata (information) about them, including their Hb Values, age, gender, and a comment focused on the Hb Value discovered during the laboratory test. It also included a photo of their conjunctiva, which was uploaded to the database for easy access. A standard high camera with a minimum resolution of 12 MP was used by certified medical laboratory personnel to take pictures of the conjunctiva of the eyes.

The dataset collection was carried out by trained and licensed medical laboratory personnel who were given week-long professional training at the Center for Research in Applied Biology, University of Energy and Natural Resources, Sunyani, Ghana, on how to capture conjunctiva of the eye images of children aged 6–59 months. The Committee for Human Research and Ethics examined and permitted (Reference number: CHRE/CA/042/22) this study before its beginning. Since participants are minors, informed consent was approved by their parents or guardians before the participants were enrolled in data collection.

With regards to taking pictures of the conjunctiva of the eyes, the lower eyelid was slightly pulled down with the use of the thumb and index finger to expose the conjunctiva of the eyes. To minimize extreme gleam effects on image quality, which might impact model performance, the cameras' spotlights were turned off when photographing the images. This method was a huge help in eliminating the impact of ambient light in the photographs.

Additionally, the cameras' spotlights were turned off when taking pictures to reduce excessive shine effects that may affect the image quality, which has a significant impact on the models' ability to recognize or classify objects. This method is a great technique to make datasets of photographs completely free of the effects of ambient light.

The features of the conjunctiva images were extracted using the triangle thresholding technique, which also obtained the ROI. The extracted Region of Interest (ROI) was then broken down into metrics by using the images' CIE L*a*b* colour space intensity value and augmented to increase the size of the dataset. The value in the CIE L*a*b* colour space represents the various metric values. Children aged 6–59 months are the primary subjects of the datasets used in the study, with 424 as anemic and 286 as non-anemic representing 60% and 40% of the total original dataset respectively.

The datasets used for this study were gathered from selected hospitals in Ghana; Komfo Anokye Teaching Hospital, Bolgatanga Regional Hospital, Kintampo Municipal Hospital, Ahmadiyya Muslim Hospital, Sunyani Municipal Hospital, Manhyia District Hospital, Ejusu

Government Hospital, SDA Hospital, Nkawie-Toase Government Hospital and Holy Family Hospital.

Phase 1. Extracting the Region of Interest

The conjunctiva is the ROI of the eye. To obtain the ROI, we first determined whether or not there was an eye in the given photograph. This was performed by comparing each collected image to a trained eye detection model that was trained on images of the eyes and conjunctiva. The eye detection model was trained and utilized in Haar Cascade forms. Once the eye was identified, it was removed from the initial image and used as the actual image for the subsequent process. The conjunctiva was extracted from the eyes and was accomplished through.

1. Finding the center of the eye

To detect the conjunctiva, first, locate the center of the eye. This was achieved by identifying the pupil of the eye, which was done with a modified version of the Hough Circle Algorithm. This algorithm recognizes circles in a given image.

2. Cropping out the unwanted part of the eye from the image

After the center of the eye was located, we then cropped out the upper layer of the image and concentrated on the lower side, which included the ROI of the conjunctiva. The average distance from the center of the eye to the lower eyelid is 3.5 mm [24]. The value (3.5 mm) was converted to its comparable value in pixels to crop the image further and focus on the exact location of the conjunctiva.

3. Extracting the Conjunctiva

After removing the undesired part of the image and finding the exact position where the conjunctiva is likely to be presented, we used image processing techniques and extracted the conjunctiva from the remaining image.

a. The images that contained noise were filtered; all coordinates of the edges discovered in the image were saved. The area of the image that remained had fewer borders, except for the lower eyelid edges.
b. After the edges were recognized, their coordinates were saved. The conjunctiva was then calculated as the area encompassed by the edges. The edges formed an imperfect, uneven circle and the entire region within that incomplete circle was assigned to the conjunctiva at this time.
c. Finally, the conjunctiva was extracted and saved as a separate image.

4. Image Augmentation of the Dataset (Images)

To increase the size of the original datasets, an image augmentation technique was applied on the original dataset. This was done because the use of inadequate data could result in overfitting [34]. The use of image or data augmentation increases the performance of machine learning models. This is because the utilization of image augmentation increases the size of the dataset to enable the models to train with huge dataset sizes or samples [30] and enhance the avoidance of overfitting [35,36].

The rotation, flipping and translation image augmentation technique was applied on the original dataset to 1520 and 1115 for anemic and non-anemic images respectively summing to 2635. The models were then trained, validated and tested on the augmented dataset before testing the deployed system on real-time datasets (human eyes).

Phase 2. Calculating the RGB and HB Levels of the Extracted ROI

Once the conjunctiva has been detected, extracted and saved as a new image, we compare the final Shape to a set of trained, precisely extracted conjunctiva data to validate and verify it is indeed a conjunctiva we extracted. The conjunctiva's Hb level would then have to be calculated. This will be accomplished by extracting the red, blue, and green pixels from the image.

Next, we computed the mean of each of the extracted hues, that is, the mean of all the red values, the mean of all the green values and the mean of all the green values.

The normal Hb values for a person without anemia according to WHO are $<$13 g/dL in adults, and 11.0 g/dL in children [37]. We used an 11.0 g/dL value, as the base value for the prediction of anemia. 11.0 g/dL was used because our dataset was gathered from children. This implies that any image whose value $\geq$ 11.00 is considered non-anemic. Similarly, values below the base standard are considered anemic.

3.2. System architecture

The architectural diagram in Fig. 1 shows distinct modules that make up the anemia detection application. The proposed system for anemia detection consists of four modules namely: The image capture module, conjunctiva detection module, anemia detection module, and historical data module. The system consists of two main components: Frontend and Backend, together with specific algorithms.

3.2.1. Front end

The front-end is a mobile application that connects to a server on which the detection, prediction, and data management modules are hosted. The app allows users to either capture a photo using their phone's camera or select an existing photo of exposed conjunctiva from their gallery for submission to the server. The app also displays test results to the user and provides communication to/from the data management module.

3.2.2. Backend

The backend is composed of the conjunctiva detection module, anemia detection module and the data management module hosted on a FastAPI server. The conjunctiva detection module ensures the availability and extraction of the conjunctiva in submitted photos and passes such data to the detection module. The anemia detection module is responsible for calculating the Hb values of the extracted conjunctiva images passed by the detection module to make the anemia classification.

The data management module handles user account details and the test results associated with them. We utilized a MongoDB database for this purpose.

3.2.3. Operational methods

In this study, we used Convolutional Neural Networks, Logistic regression, Classification, and Gaussian Blur for anemia detection and deployed the models to a smartphone application that was hosted on a FastAPI server. The flow diagrams in Figs. 2 and 3 illustrate how the proposed system works.

A cross-platform mobile application that runs on the Android and iOS operating systems was created to React Native in the Visual Studio Code Integrated Development Environment (VS Code IDE). This app operates as the user interface or front through which the user will communicate with the Anemia Detection Model. The user launches the app and uses their phone's camera to take a picture of the eye. The image is transferred

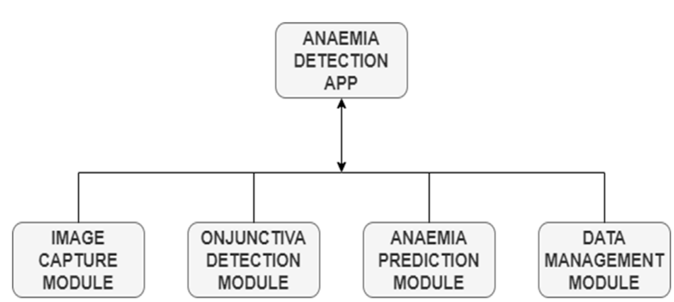


Fig. 1. Proposed system architecture.

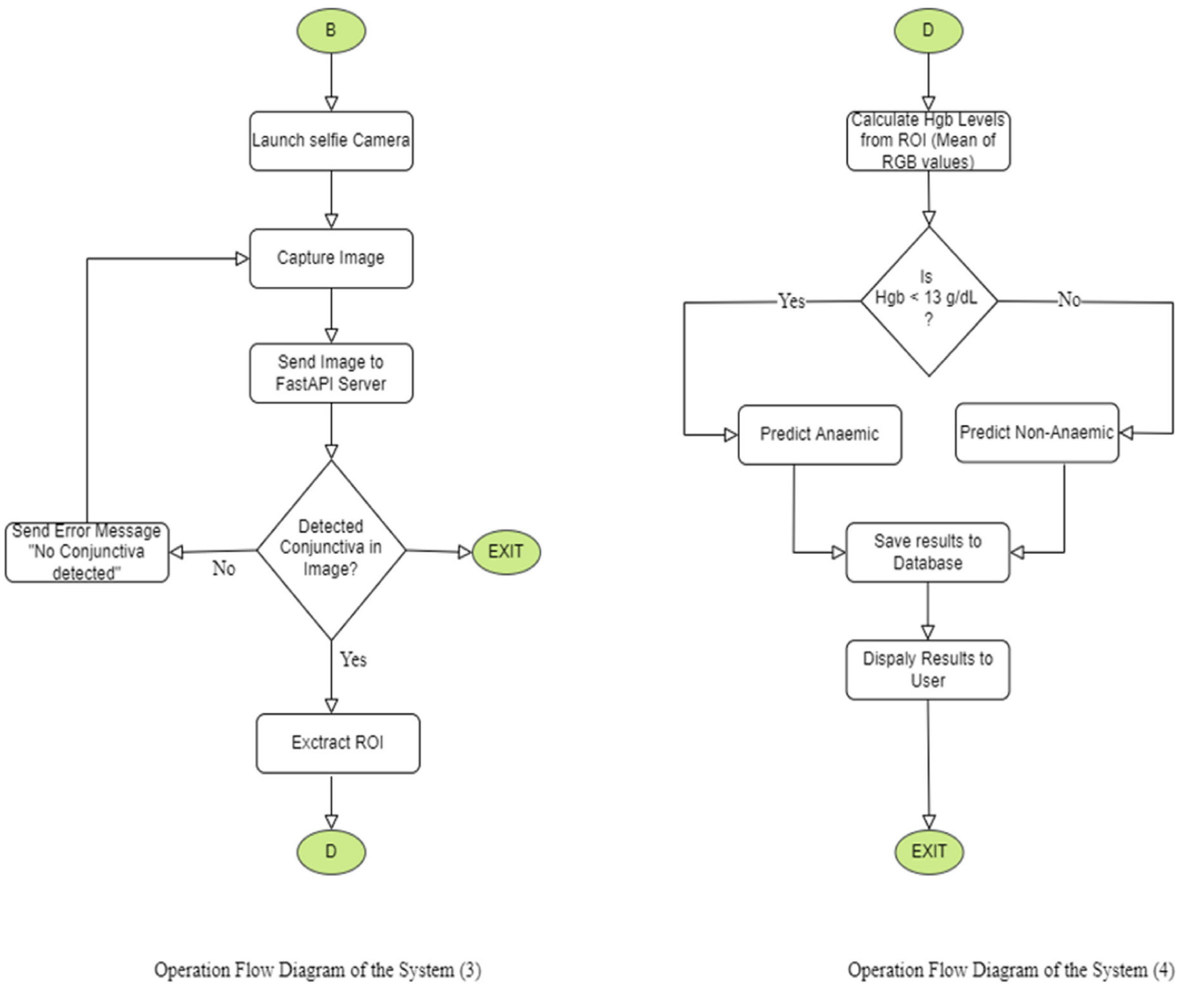


Fig. 2. Operational flow diagram of the system.

to the FastAPI server, which executes our analysis model over the phone's internet connection. The image processing occurs on a Python-based FastAPI server.

4. Results and analysis

4.1. Conjunctiva detection

The *YOLOv5* algorithm was used to automatically detect and extract the conjunctiva of eye images and create the ROI. This model was trained on *Google Collab* with 104 conjunctiva-exposed images and 20 non-eye images.

The model was trained, validated, and tested with 1844, 264, and 527 images of the eyes respectively. The training, validation, and testing dataset (images of the eyes) represented 70%, 10%, and 20% of the total (augmented) dataset used. Thereafter, we tested the models on 153 eye images, and 48 non-eye images (real-time data). The conjunctiva detection and validation results are summarized in Fig. 4.

The model correctly identified and successfully extracted 150 out of the 153 test data (True Positives = 150) and incorrectly identified 4 as non-conjunctiva images (False Negatives = 3). The system also correctly identified all 48 images without the eye as images without the conjunctiva (True Negatives = 48) and did not incorrectly label a non-eye image to contain the conjunctiva (False Positive = 0). 69 of the 71 images used for validation were positive. The conjunctiva detection accuracy of the model is depicted in Fig. 5.

4.2. Noise reduction

Using the Gaussian Blur algorithm in Python's OpenCV library, a considerable amount of blurring was applied to the images as a form of noise reduction in preparation for conjunctiva extraction. This was done to smoothen the images and eliminate some of the irregularities such as exposure and camera flashes.

4.3. RGB extraction and mean value calculation

Once the conjunctiva is successfully extracted and saved as a new image, the Red, Green, and Blue (RGB) components of the image are extracted next, and their mean values are calculated. Fig. 6 represents the extraction of RGB components.

The best detected result was reached by integrating three-component features in total, namely a*, b*, and the G value derived from the RGB component photos, mapping the RGB to CIE L*a*b* colour space (RGB→CIELAB). The L and RGB component values were retained for filtering the incoming data. The L and RGB component values were retained for filtering the incoming data. The mean values of the characteristics a*, b*, and G were carefully calculated considering only the image pixels with min $<$ L and min $< R = G = B <$ max.

This last filtering guarantees that image pixels that are abnormally dark or bright are deleted; therefore, the technique shown here takes into account only pixels that enable an accurate pallor evaluation of the photos [15].

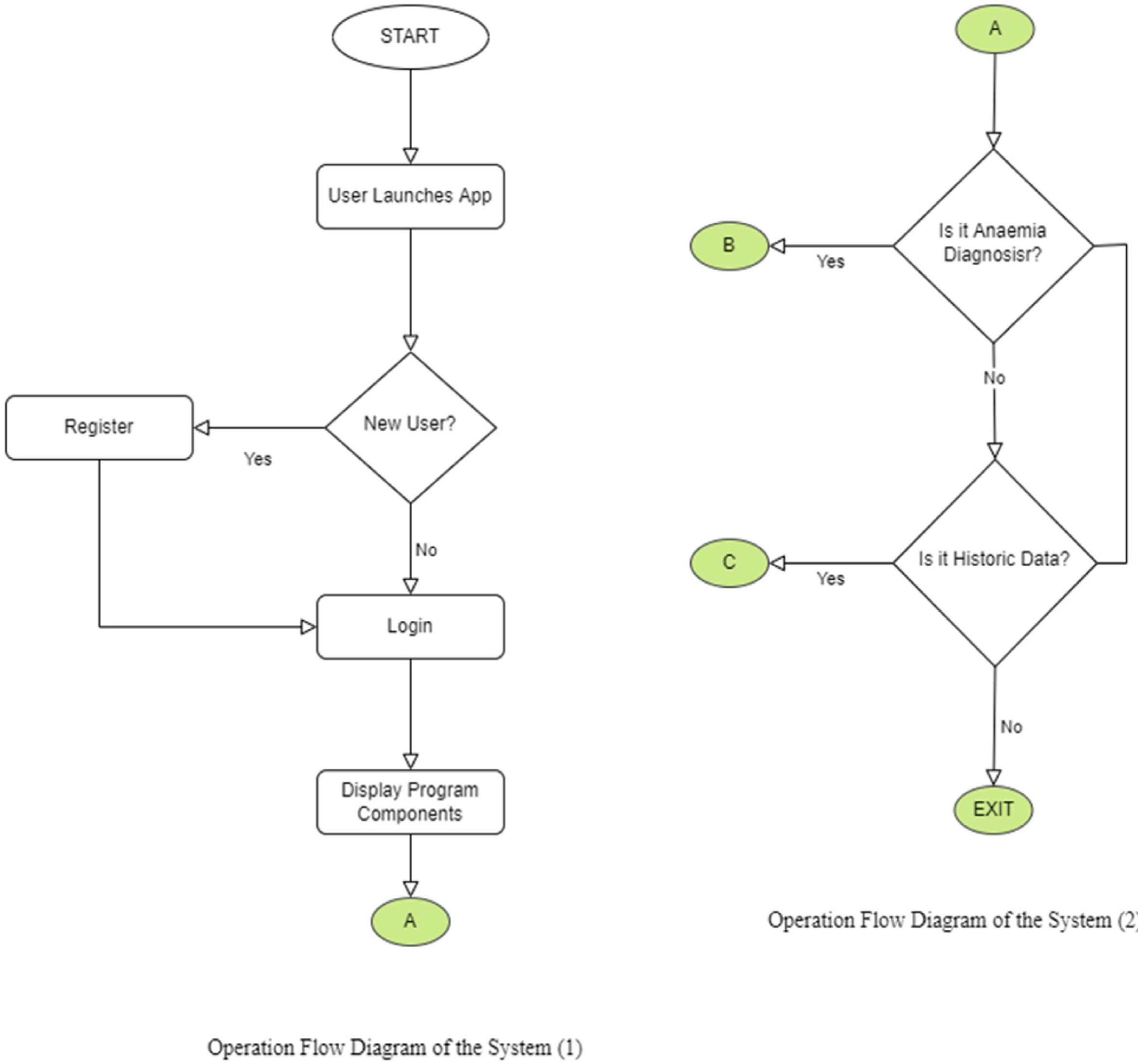


Fig. 3. Operational flow diagram of the system. This diagram is a continuation of the flow to the end.

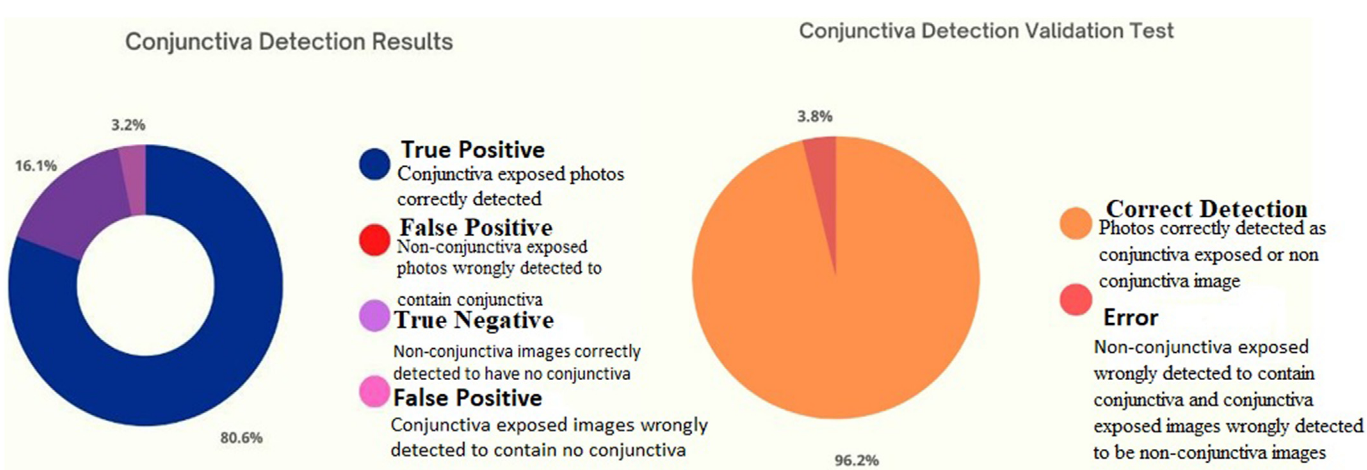


Fig. 4. Conjunctiva detection and validation results.

Once that is done, the redness of the conjunctiva (which is the image at this stage) is calculated. The results of this calculation give us the Hb value of the image in a range between 0 and 1. This calculation is achieved using the formula in equation (1) as adapted by Ref. [24];

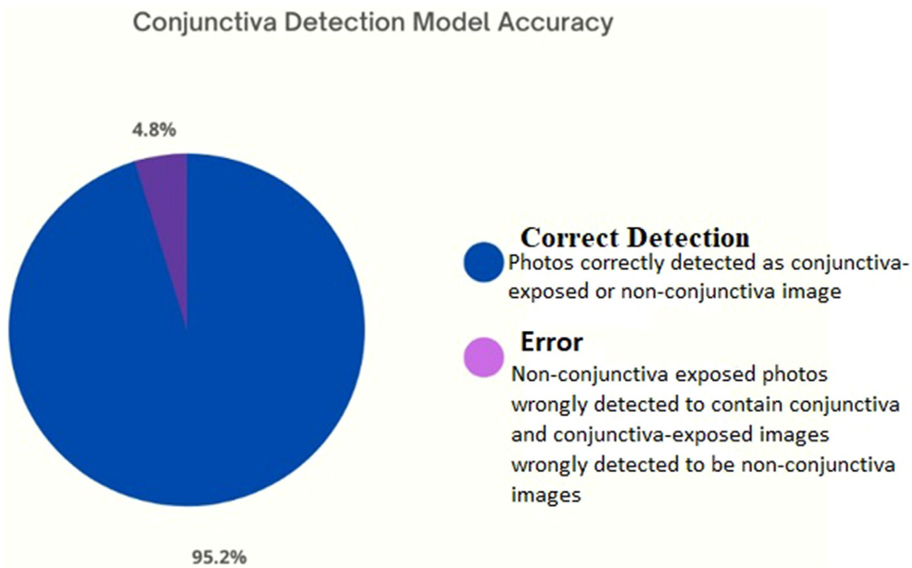


Fig. 5. Conjunctiva detection model accuracy. The conjunctiva detection model achieved an accuracy of 95.20%.

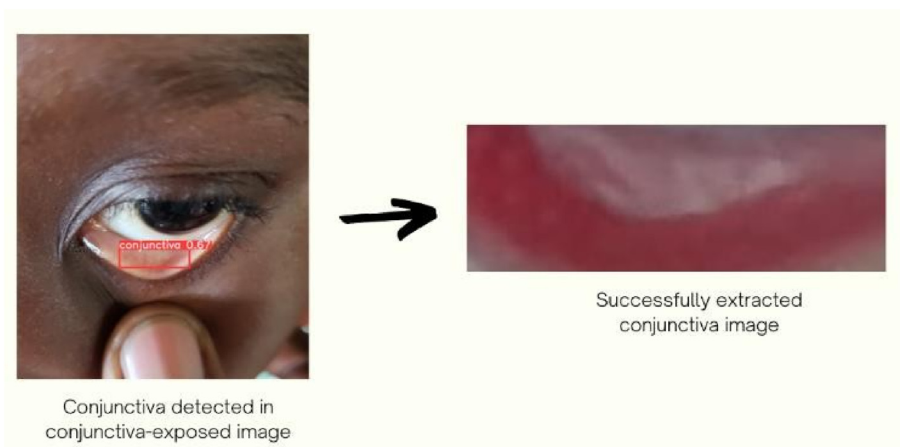


Fig. 6. Conjunctiva extracted.

$$Hb = \frac{e - 1 : 922 + 0 : 206{*}r - 0 : 241{*}g + 0 : 012{*}b}{1 + e - 1 : 922 + 0 : 206{*}r - 0 : 241{*}g + 0 : 012{*}b} \quad (1)$$

where r is the mean of the red pixels, g is the mean of the green pixels, and b is the mean of the blue pixel.

4.4. Anemia classification

To classify the result as anemic or non-anemic, the Hb value obtained in the range 0–1 was converted to our threshold range with values 7 to 15 where 7 is the minimum range and 15 is the maximum range. This was achieved by using equation (2):

$$f(t) = \frac{c + (d - c){*}(t - a)}{(b - a)} \quad (2)$$

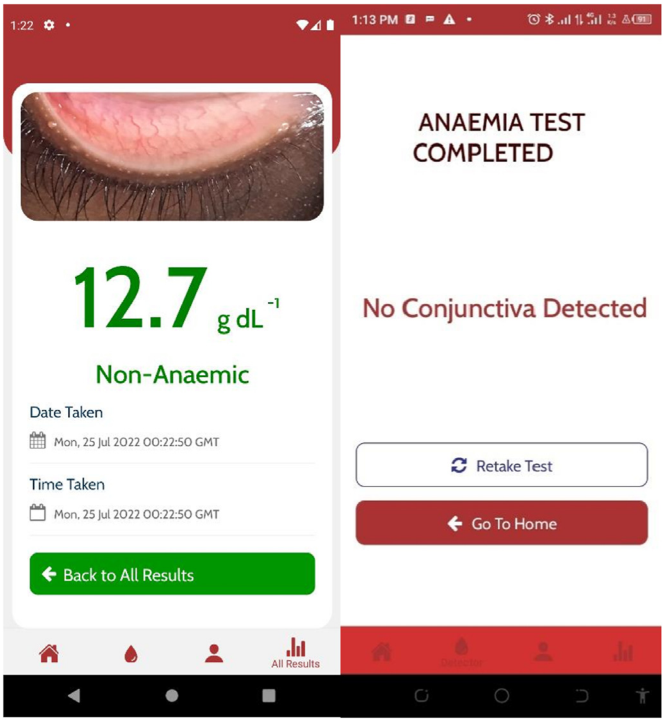
Here we convert the initial range (a-b) to our new threshold range (c-d). This gives us an Hb value in the range of 7–15 g/dL. Comparing this value to the standard WHO value (<11 g/dL) for anemia, we classified the result as anemic or non-anemic.

The calculated Hb value is saved with its classification, interpretation, and operation time stamps in a database for future access and reference. A response is then generated to the user indicating their Hb level and it corresponding anemic status in Fig. 7 A. In an event that conjunctiva is not detected in the photo submitted, an error message is displayed instead in Fig. 7 B and Fig. 8 gives a pictorial view of the steps to use the app.

4.5. Using the application

The steps outlined below show how the application can be used with ease.

1. Installation and launch of the application.
2. Login if you already have an account.
3. Sign up if you have not already registered.
4. On the main page select “Anemia Detector” after successfully signing in.
5. Take a picture of your conjunctiva using your phone camera or select an existing facial photo with exposed conjunctiva from your gallery.
6. Click on “Submit” and wait for the results.
7. Retake the test or go back to the start page.



A B

Fig. 7. Successful detection (A), unsuccessful detection and prediction (B).

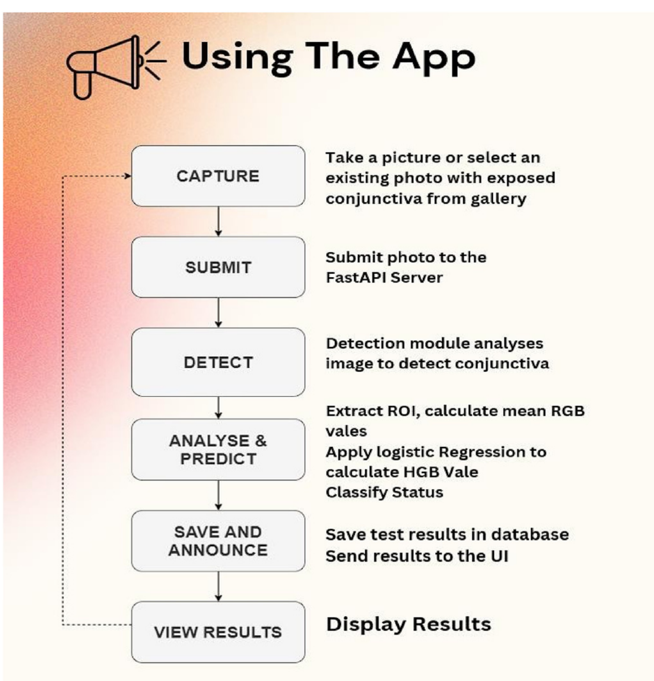


Fig. 8. Steps to use the application. This represents the pictorial view of the steps used to utilize the application.

4.6. Evaluation metrics

In post-implementation testing, we used a total of 100 subjects for validation. These subjects were clinically tested and grouped into anemic and non-anemic categories based on their Hb values. We deliberately made sure to have an equal number in each group to neutralize the testing results. So, for each group, there were 50 subjects. After taking the conjunctiva photo of each patient and submitting for anemia prediction twice using the app, the following results were obtained:

4.6.1. Accuracy

Our application was able to correctly predict 45 out of 50 images as anemic and 47 out of 50 as non-anemic in the first test. In the second test, 47 out of 50 images were correctly predicted as anemic and 46 as non-anemic. This implies that our app achieved an accuracy of 92% on the first test and 93% on the second test. Overall, this app correctly predicted anemia status with an accuracy of 92.5%.

4.6.2. Sensitivity

The test results also showed that the app was fairly sensitive to anemic photos (presence of anemia). Specifically, it correctly detected 44 out of 50 (88%) anemic conditions in the first test, and 46 out of 50 (92%) in the second, totaling 90%.

4.6.3. Specificity

Referring to the detection rate of non-anemic photos, the specificity was determined to be 95%. In the first test, 48 out of 50 non-anemic photos were corrected and detected, and 47 out of 50 were detected in the second, summing up to 95 out of 100.

5. Operational Time

The operational time here refers to how long it takes for the model to run analysis on an image and provide results to the user right after the 'Submit' button is clicked. Results from our tests indicated an average runtime of 45 s. Table 2 give details of the various performance of the models, while Table 3 compare the performance of related studies and Fig. 9 demonstrate the anemia detection app.

The best recorded time was 7 s, and the longest was 1 min and 10 s. Figs. 10 and 11 represent a summary of the accuracy results and the evaluation metrics of the developed anemia detection application, and Fig. 12 gives a graphical representation of related works performance andd method used.

5.1. Discussion

A sensitivity of 65% implies that the app correctly identifies truly anemic images fairly. While this figure is low compared to similar

Table 2
Summary of the smartphone application performance when tested on real-time data.

	First Test	Second Test	Average Performance
Accuracy	92%	93%	92.5%
Sensitivity	88%	92%	90%
Specificity	96%	94%	95%
Operational Time	80 s	45 s	5 Seconds

Table 3
Summary of related works performance and method used.

Reference	Algorithm Used	Results
Tamir et al. [21]	SVM	78.90%
Noor et al. [29]	k-NN, SVM, and Decision Tree	73.91% (k-NN)
Irum et al. [38]	SVM	78.90%
Delgado-Rivera et al. [39]	CNN	77.58%
Peker et al. [40]	SVM	64.66%
Getaneh et al. [41]	HemoCue	58%

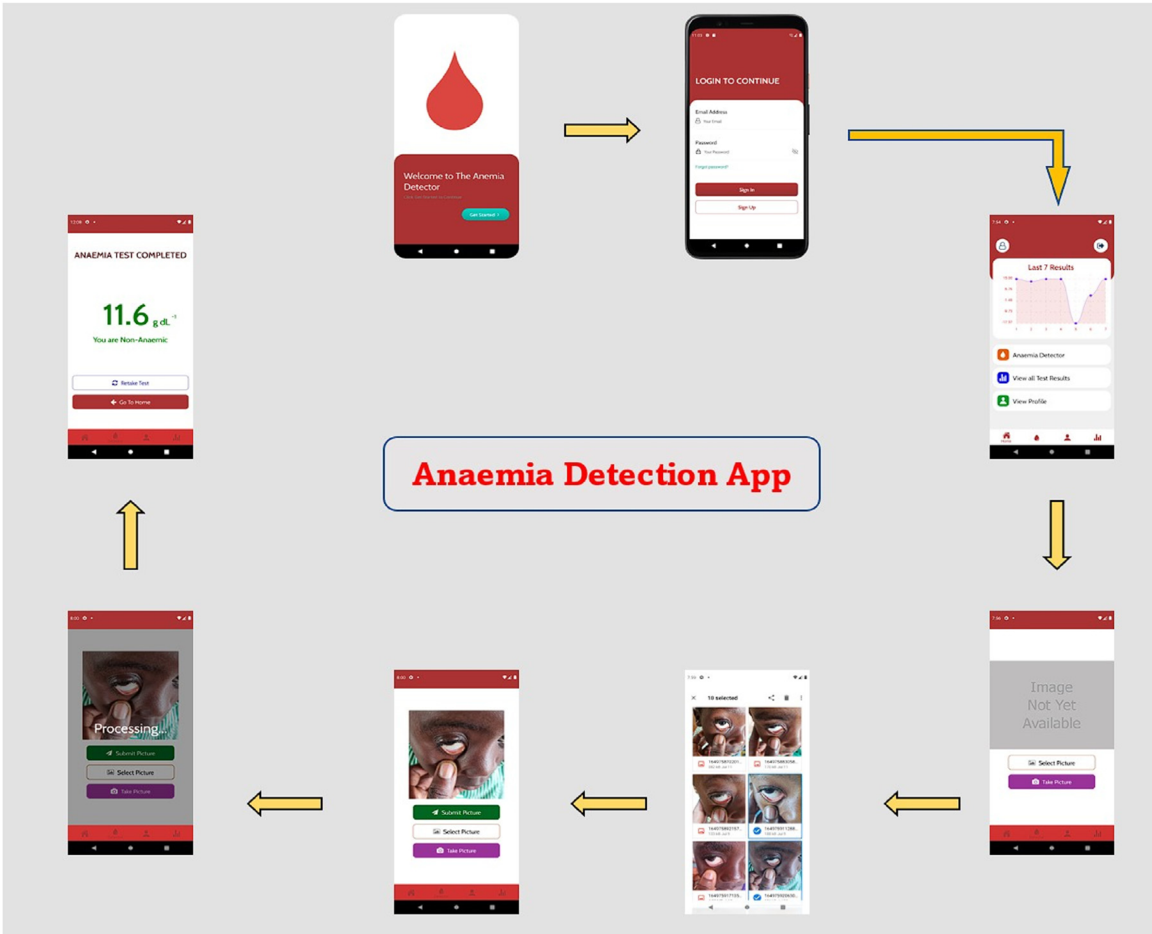


Fig. 9. Demonstration of the anemia detection application. Additionally, this is a sample of the results of how the application works with practical demonstrations. This provides easy-to-use steps and processes to determine the classification of anemia based on the conjunctiva images.

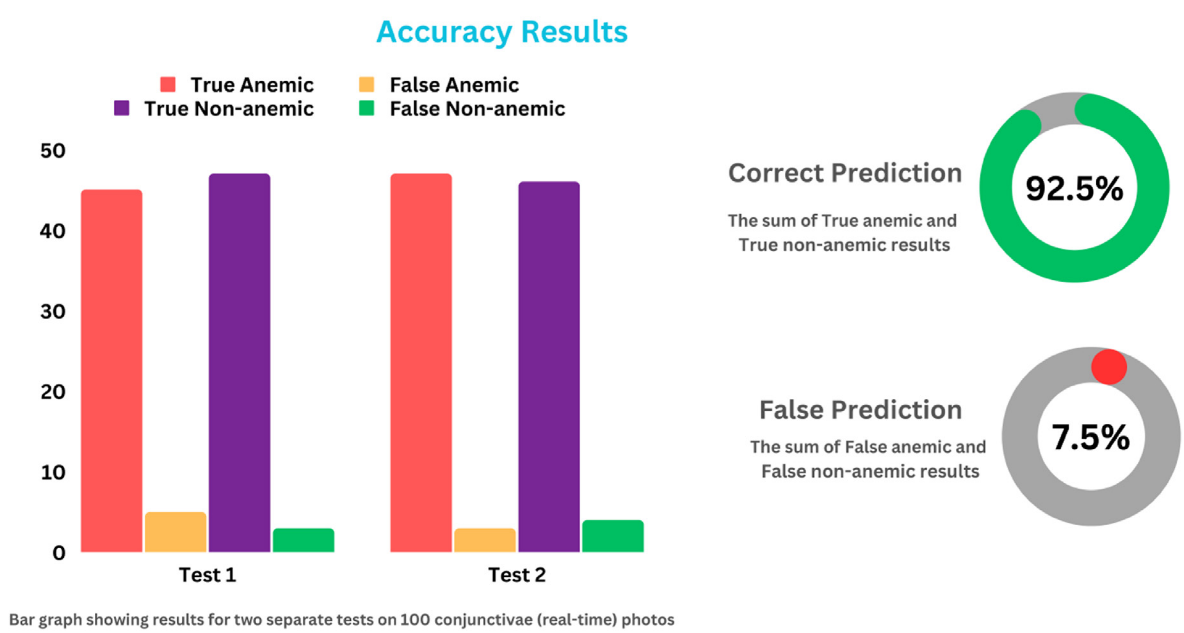


Fig. 10. Summary of the results of the anemia detection application accuracy.

projects such as ShEMO (89%) and HemaApp (83.3%) it outperforms others such as that [20]. Having a low sensitivity reduces the number of possible false positives, which could cause panic in patients and lead to wrong predictions by physicians. On the other hand, there are high chances of missing anemic situations, especially in cases where the Hb value is close to the base Hb value (11.5 g/dL).

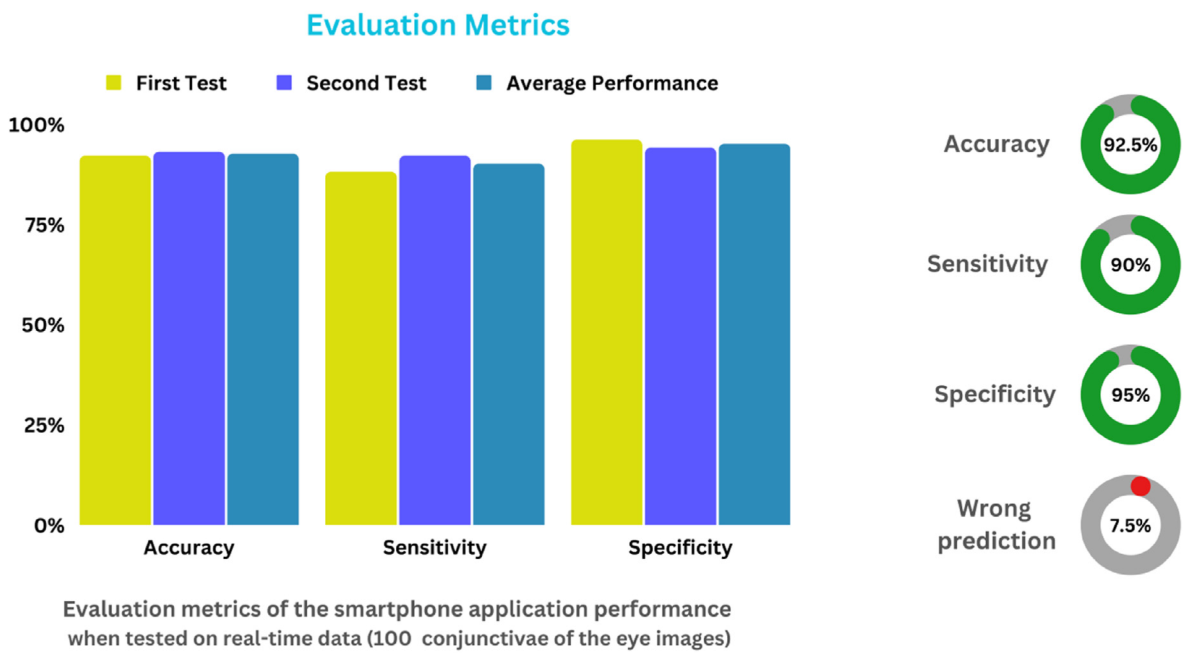


Fig. 11. Summary of the evaluation metrics of the anemia detection application.

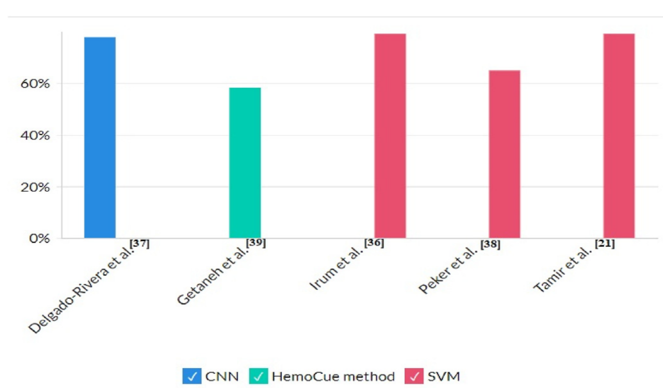


Fig. 12. Graphical representation of related works performance and method used as indicated in Table 3.

In the study by Tamir et al. [21], the SVM achieved an accuracy of 78.90% which was higher than the result of Noor et al. [29] where the k-NN performed better than the SVM and the Decision Tree with an accuracy of 73.91%. Moreover, Delgado-Rivera et al. [39] achieved an accuracy of 77.58% with the application of CNN, as compared to Peker et al. [40] and Getaneh et al. [41] which achieved an accuracy of 64.66% and 58% when the SVM and the HemoCue were applied respectively applied in the proposed method.

However, the study by Refs. [21,29,38–41] developed a proposed model for detecting anemia with their respective study, while this study we developed the proposed models and implemented them on a smartphone application, which is available on google play store platform to be accessed by all for the detection of anemia.

The proposed system achieved a significant result of 92.50% (accuracy) with 5 s of operational or running time when tested on real-time data. This is a massive approach as compared to the related studies' results obtained in Table 3 whereby the proposed models were not tested on real-time data (images) to justify the performance of the models outside the trained, validated and tested datasets.

In terms of specificity, this app excels at detecting non-anemic images. This presents low chances of reporting false negatives (i.e., Predicting truly anemic cases as non-anemic). Having good specificity ensures that anemic conditions are not wrongly concealed. Using this metric, this app outshines similar projects such as Espinoza et al. [14] which recorded a specificity of 36.03%.

One of the unique features of this app is its data management module which is non-existent in similar projects like HemaApp. This allows users to keep track of all their test results and conveniently displays the last seven results in a graph on the home screen of the application after signing in. This feature is very important, particularly for people with chronic anemia who need to monitor their Hb levels regularly. These results can be downloaded and shared with a physician for expert advice and guidance.

This module also allows users to switch devices and still have access to previous test results attached to their user accounts. Although the internet speed of the user could affect the evaluation time of the app, the app boasts an average runtime of 50 s. This means that, unlike a hemoglobinometer test which takes more than 15 min [42]. With the app developed by Hermoza et al. [42], one can test the anemic status and have your results displayed on your smartphone in less than a minute. Although the use of smartphones with a poor camera (below 5 MB) may affect the quality of images and can affect the detection and prediction efficiency of our model, the constant improvements of smartphone cameras mitigate this limitation.

5.2. Conclusion

In this study, we used image processing and machine learning models to develop a mobile phone application that quantitively analyses the pallor in conjunctiva images (photos) to predict anemia with an accuracy of 92.50%, a sensitivity of 90% and a specificity of 95% with a runtime of 50 s after being tested on 100 conjunctiva images (real-time images).

The results indicated that this app is a good replacement for invasive anemia diagnosis (detection), particularly in the African region where anemia is more common. Although the use of smartphones with poor cameras (below 5 MB) may affect the quality of images and subsequently the effectiveness of our model's detection and prediction, this limitation is mitigated by ongoing improvements to smartphone cameras.

The testing time of the application may be impacted by a poor internet connection, but with a reliable network, one can anticipate receiving test results in under a minute, with the potential to accelerate to as little as 5 s.

We intend to make the software operational offline in the future to enable its accessibility in remote areas where network quality may be poor, after the use of more model training with more data. The anemia detection application has been hosted on Google Play Store for Android users to access the application without a fee.

5.3. Future works and recommendations

Future works would look into upgrading the application onto App-store for iOS users to also access the application and provide free training for clinical purposes. Going forward after further training and testing of our prediction model. In the future, we plan to make the software offline to enable access in remote locations where the network quality may be poor. This will happen after more model training with more data.

Authors' contribution

Justice Williams Asare: Conceptualization, Data curation, Methodology, Writing – review & editing. Peter Appiahene: Conceptualization, Data curation, Formal analysis, Supervision. Enoch Justice Arthur: Writing- Original draft preparation, Methodology, Software. Stephen Korankye: Writing- Original draft preparation, Methodology, Software. Stephen Afrifa: Methodology, Writing – review & editing, Project administration. Emmanuel Timmy Donkoh: Conceptualization, Data curation, Supervision. All the authors read and approved the article.

Funding statement

The research received no external funds.

Declaration of competing interest

The authors declare that there is no conflict of interest in this study.All the authors read and approved the article.

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